Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: FINORA
b. Team leader name: Dhilip S E
c. Problem Statement: BFSI: Intelligent Risk, Fraud & Financial Experiences.
   Product: CASCADE, a cash-flow risk radar for people with irregular income

Brief about the idea

The problem

A delayed client payment, a loan EMI and a card bill can land in the same week. Banking apps show the balance today. They do not show the combination that pushes the account negative three weeks from now.

What we built

CASCADE reads a user's income, loans, cards and bills, projects 60 days of cash flow, and uses Gemini to propose compound what-if scenarios the user would not think to ask about. A deterministic engine, not the AI, simulates each scenario with exact integer arithmetic. Gemini then explains the verified result in plain language and suggests what could help.

Built for

Freelancers, gig workers and small-business owners in India (INR first). Informational only; not financial advice.

60 days

projected, day by day

2+

interacting events per scenario

0

numbers invented by the AI

Opportunities

How different is it from existing ideas?

- Budgeting apps record the past; CASCADE looks forward and tests combinations of events.
- Bank apps show a balance; CASCADE shows the day the balance goes negative and why.
- Chatbots generate numbers; here the AI never produces a figure. The engine does.

How will it solve the problem?

- Gemini proposes 2+ event scenarios from the user's own records.
- Every scenario is validated against real record ids, then simulated exactly.
- Interventions (shift a due date, top up the reserve, cut spend) are compared against the baseline.

USP of the proposed solution

- Verifiable: every amount traces to a record and a rule version (engine-1).
- Honest: unknowns, like a missing grace period, are listed, not guessed.
- Safe: untrusted-data prompts, schema-checked output, no probabilities, no product advice.

List of features offered by the solution

Sign-in and demo mode

Firebase Auth: email, Google, anonymous. One-tap synthetic demo persona.

Financial records

Income, loans, cards, rent, insurance, savings. Frequencies, grace days, variable rates.

60-day cash-flow view

Daily, weekly and monthly projection, lowest-balance day and shortfall dates.

AI scenario discovery

Gemini proposes compound scenarios; each is validated and simulated.

Simulate and compare

Deterministic engine. Baseline vs alternative; which input moves the result most.

Interventions

What could help: shift an event, use grace periods, top up reserve, cut spending.

Assistant

Ask in plain words: 'What if Client A pays 10 days late?' Asks back when details are missing.

Document extraction

Gemini reads a loan or card document: value, page, excerpt, conflicts, injection flag.

Alerts and push

Threshold alerts with quiet hours and de-duplication; FCM push.

Manual events, undo

Add, move or remove cash events; undo the last change.

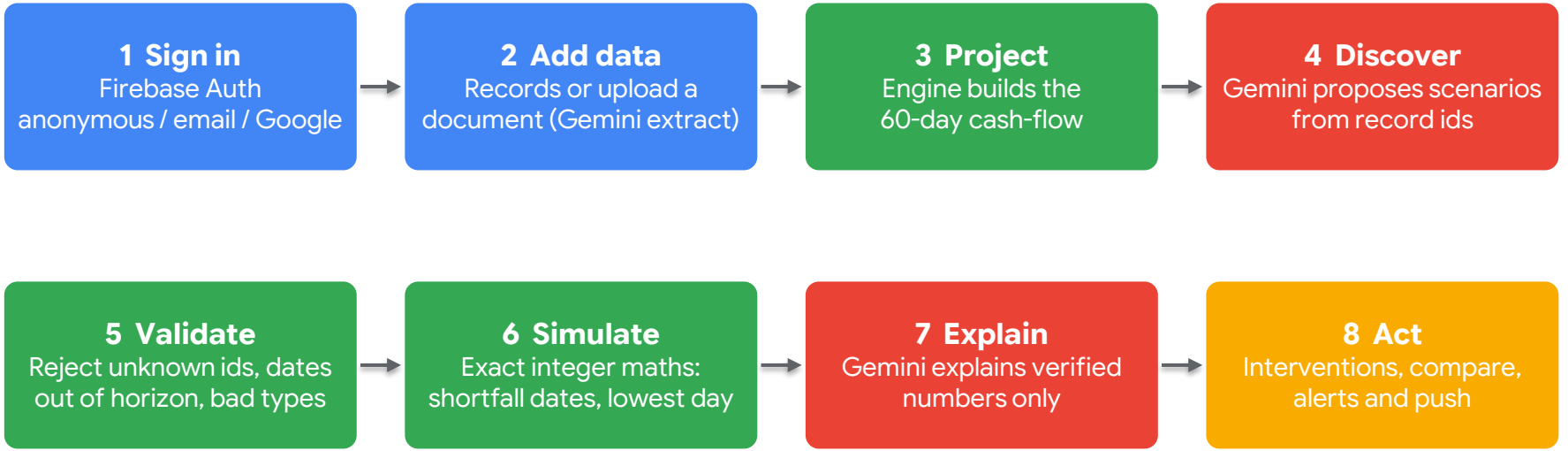
Privacy controls

Export all data, delete account and data, per-user isolation.

Not advice, by design

Every screen labels outputs as possibilities with known and unknown inputs.

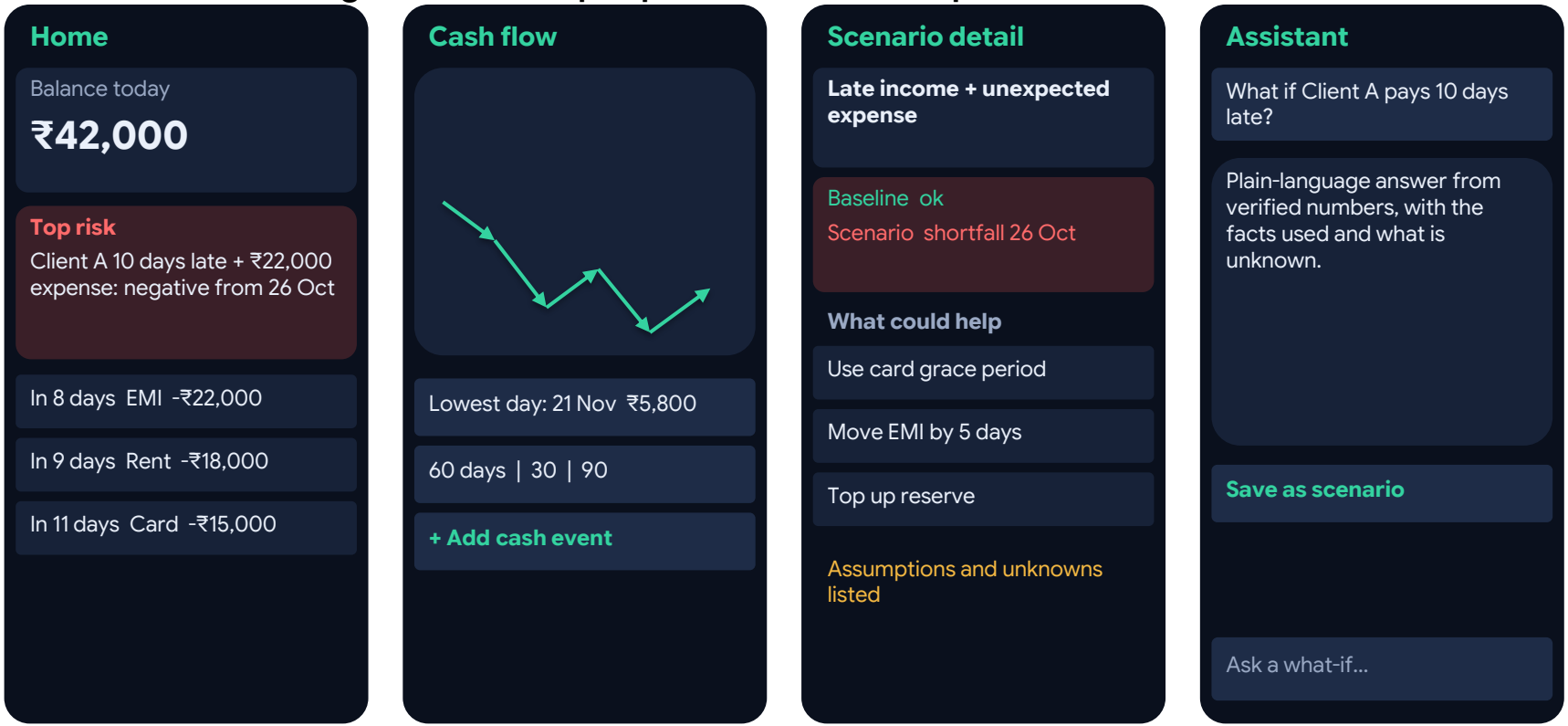
Process flow diagram or Use-case diagram



Use case: Asha, a freelancer, has ₹42,000 and a ₹60,000 client payment due in 6 days, with a loan EMI, rent and card bill right after. CASCADE finds that a 10-day client delay plus a ₹22,000 surprise expense turns the account negative from 26 Oct, lowest -₹64,200 on 2 Nov, then shows what could help.

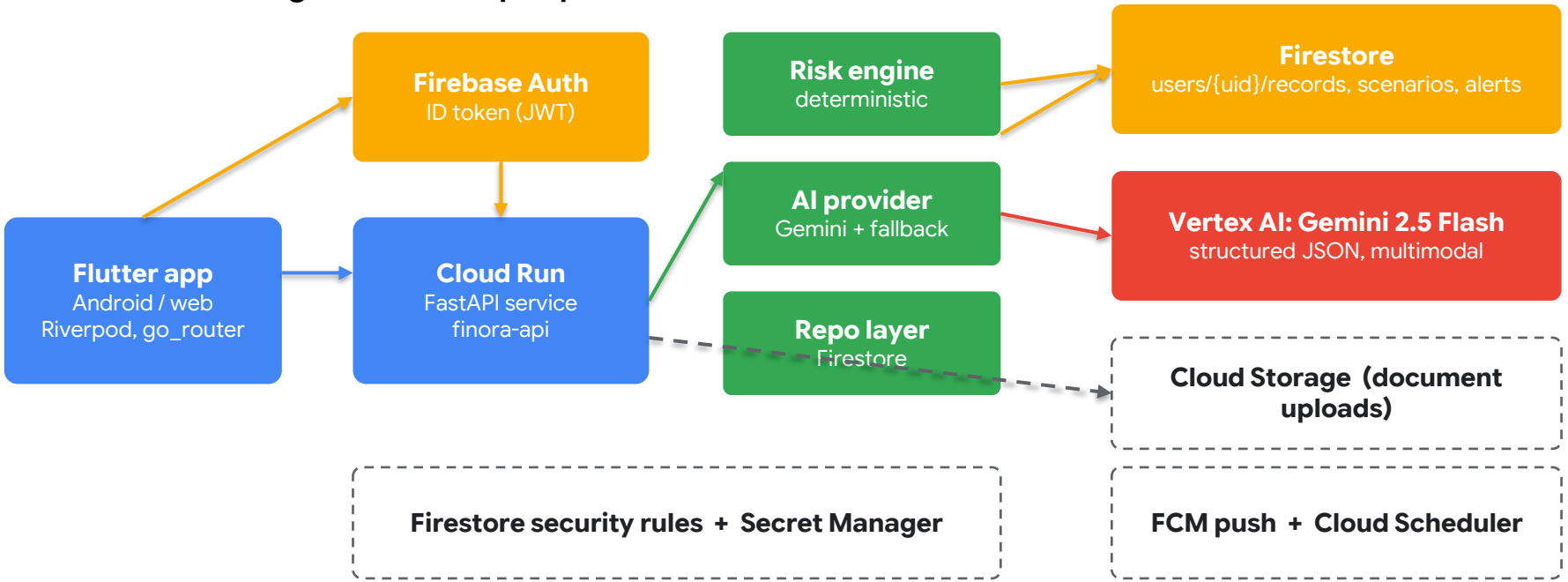
Rule: the AI proposes and explains. The engine decides every number.

Wireframes/Mock diagrams of the proposed solution (optional)



Wireframes of the implemented Flutter screens (dark theme: emerald = healthy, amber = uncertain, red = projected negative).

Architecture diagram of the proposed solution



Solid = deployed and tested. Dashed = implemented in the code, enabled per environment.

The API holds the Gemini credentials (Vertex AI uses the Cloud Run service identity). The app never calls Gemini directly.

Technologies to be used in the solution

Google Cloud

- Gemini 2.5 Flash on Vertex AI (google-genai SDK)
- Cloud Run (container, scale to zero)
- Cloud Build + Artifact Registry
- Cloud Storage, Cloud Scheduler, Secret Manager

Firebase

- Cloud Firestore (data + security rules)
- Firebase Authentication (anonymous, email, Google)
- Firebase Cloud Messaging
- Firebase App Check (Play Integrity)

Backend

- Python 3.11, FastAPI, Pydantic v2
- Integer minor-unit money, half-up rounding
- Rate limiting, typed error envelope
- pytest: 51 tests

Mobile / web

- Flutter (Dart)
- Riverpod, go_router
- fl_chart for the projection
- flutter_secure_storage

Estimated implementation cost (optional)

Component	Pricing basis	Prototype estimate
Cloud Run	Per request and CPU time; scales to zero	Within free tier at demo traffic
Firestore	Reads, writes, storage	Within free tier at demo traffic
Gemini 2.5 Flash (Vertex)	Per token; about 3 short calls per scenario run	A few cents per 100 sessions
Firebase Auth / FCM / App Check	Free tier covers anonymous, email, push	No cost
Cloud Build / Artifact Registry	Per build minute, storage	Under US$1 per month

Estimated prototype running cost: under US$10 per month at demo traffic.

These are estimates from published pricing models, not measured bills. Cost per active user is dominated by Gemini tokens, and the app calls Gemini only on discovery, assistant, explanation and document extraction, never on page views.

Snapshots of the prototype

Live output for the synthetic demo persona (60-day horizon, INR). Values below are the engine's response, not mock data.

₹42,000
Opening balance, 18 Oct

₹5,800
Lowest projected balance, 21 Nov

₹32,600
Closing balance, day 60

14
Cash events in horizon

Scenario found by discovery

Late income plus an unexpected expense: Client A (design retainer) is 10 days late and an unplanned ₹22,000 expense, equal to the largest single obligation, lands on 26 Oct. Result: first shortfall 26 Oct, lowest balance -₹64,200 on 2 Nov, 39 days negative (the 10-day delay alone: -₹42,200, 15 days negative).

Missing information reported: "Grace period for 'Personal loan EMI' is unknown."

Labels returned with every result: numbers = simulated_by_engine; hypotheses = AI or rule-based demo fallback, shown explicitly.

Screens: welcome, auth, profile setup, home, cash flow, records, documents, scenarios, scenario detail, assistant, alerts, settings.

Prototype Performance report/Benchmarking

51 / 51

backend tests passing (engine, API, Gemini adapter)

5.5 ms

median, GET /cashflow, 60-day horizon

9.6 ms

median, scenario discovery incl. simulation (fallback provider)

0

AI-generated numbers: all amounts come from the engine

Method

- Measured in-process with the FastAPI test client, in-memory repository, synthetic demo data, 30 runs (cash flow) and 10 runs (discovery). It excludes network, Firestore and Gemini latency, so it shows the engine cost, not end-to-end latency.
- Determinism: same input gives the same output (no randomness, no clock access in the engine); money is integer minor units.
- Safety checks in tests: dev auth refused when ENV=prod, schema-invalid model output retried then rejected, prompt-injection flag on documents.

Additional Details/Future Development (if any)

Real data in

Account Aggregator / bank feeds and UPI statements instead of manual entry.

Fraud signals

Flag unusual debits and mule-style inflows against the user's own pattern.

Reach

Hindi and regional languages; voice what-if; WhatsApp alerts.

Trust

Per-field source links in the UI; audit trail export; App Check on web.

Lender view

Early-warning API for lenders to offer restructuring before a missed EMI.

Calibration

Back-test discovery against users' actual outcomes to rank scenarios.

Provide links to your:

GitHub Public Repository

https://github.com/DHILIP-S-E/FINORA

Demo Video Link (3 Minutes)

VIDEO_LINK

Final Product Link

https://finora-api-510528501361.asia-south1.run.app/docs (live API + Swagger UI)

Responsible AI and security

AI cannot invent numbers

Prompts forbid new amounts, dates and ids. Output is schema-validated JSON; unknown record ids are rejected.

Untrusted data stays data

Record names, notes and document text are wrapped as data. Documents are scanned for instructions to the AI.

No false certainty

No probabilities, no 'will happen'. Results are possibilities, with known inputs and unknowns listed.

Access control

Firebase ID tokens verified server-side; per-user Firestore paths and rules; rate limits on AI routes.

Secrets and identity

Gemini via the Cloud Run service identity (Vertex AI); no keys in the app; dev auth blocked in production.

User control

Export or delete all data; every AI output is labelled as AI or as the demo fallback.

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Thank you

